

Smart Recycle Bin

Group 7

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Agenda & PRESENTATION ORDER

- Introduction (with Subspace) - Diparshan
- Data Source - Fari
- Persona - Sergio
- AI Intents - Hsi-Teng Liu
- Prioritization - Chia-Wei
- To-Be Scenario Map - Beekash
- Prototype 1 - Gavriel
- Prototype 2 (and Conclusion) - Callum

Introduction

The Future of Waste Management: Why Change Is Urgent

- Traditional waste collection is inefficient and outdated
- Fixed collection routes cause fuel waste, pollution, and overflow
- Improper sorting leads to low recycling rates and regulatory fines
- Manual tracking is labor-intensive and error-prone
- Growing demand for smart, sustainable urban solutions



AI-Powered Smart Bin

- Combines image classification + sensor data for waste recognition
- Enables real-time alerts and dynamic route planning
- Improves recycling accuracy and reduces contamination
- Delivers data-driven reports to support management decisions
- Target users: Restaurants, urban waste departments, and smart cities



Data Source:

Public & have:

- **Toronto Waste Management Data (Solid Waste Tonnages):**
Provides solid waste tonnage data crucial for understanding waste trends in urban areas
- **OpenAI Environment & Sustainability Datasets:**
Offers insights into broader environmental impacts and sustainable practices.

Data Source:

Public & Need:

- Real-time waste generation statistics from Toronto Open Data (if available)
- Waste generation trends from GODAN (Global Open Data for Agriculture and Nutrition)

Private & Need:

- User interaction data with smart bin (how/when it's used)
- Data on types of non-recyclable items detected
- Waste management in the U.S | Statista

Data Source:

User & Nice to have:

- User feedback data

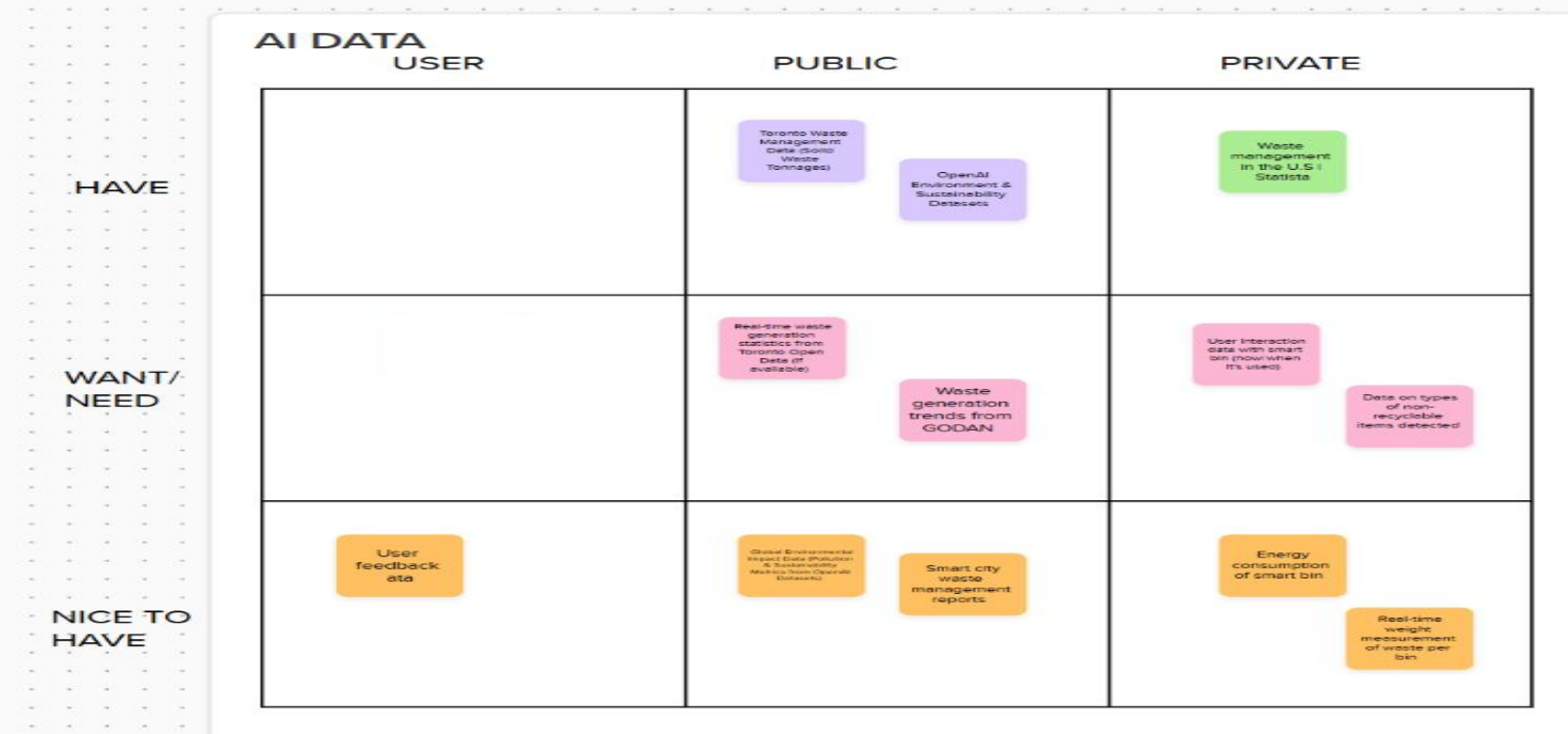
Public & Nice to have:

- Global Environmental Impact Data (Pollution & Sustainability Metrics from OpenAI Datasets)
- Smart city waste management reports

Private & Nice to have:

- Energy consumption of smart bin
- Real-time weight measurement of waste per bin

Data Source:

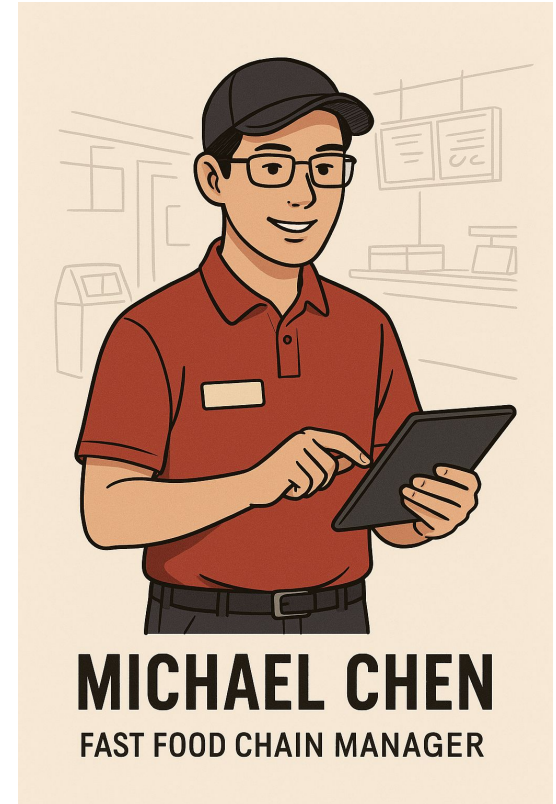


Target Audience



Personas

- Michael Chen
 - a. Age: 35
 - b. Work Experience: Over 10 years in restaurant management
- Need and Pain Point
 - a. Wants to minimize waste sorting errors to avoid fines or customer complaints
 - b. Hopes to have real-time data and reports for operational management and reporting to superiors
 - c. Worries about initial costs and maintenance difficulties of new system adoption



AI Intents: Recommend with Confidence

- Correct waste classification.
- Classifications according to the type of bins which should be categorized to at list 3 principal spots: recyclable, organic, waste.
- Direct to waste bin when the classification is of low confidence.
- False classifications can be corrected via user feedback on screen for further training.
- A primary model for each industry is trained while strategically utilizing models from other industries.



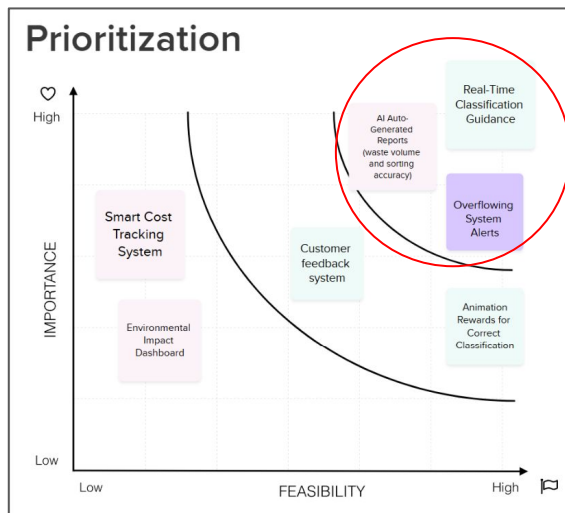
AI Intents: Enrich Your Interactions

- Direct people where to drop the trash according to classification result.
- Non-disruptive pleasant sound when placed in correct bin
- Warnings to management of problems such as malfunctioning and overflowing.
- Customer reward programs with participating industries.
- Statistics reports to the management and headquarter.

Smart Bin Feature Prioritization

Goal

- Select features using insights from **Big Idea Vignettes**
- Focus on features with **high importance and high feasibility** that satisfy the Hills.



Big Idea Vignettes

BIG IDEA VIGNETTES

Classification Experience Optimization

Real-Time Classification Guidance:
As a customer approaches the bin, the smart bin's screen displays visual instructions on where to place each type of waste based on real-time camera recognition.

Animation Rewards for Correct Classification:
After a user correctly sorts an item, the smart bin immediately plays a short, cheerful animation as positive feedback.

Problem Statement:
Michael Chen struggles with waste bins frequently overflowing and waste classification during busy hours. These issues cause hygiene problems, customer dissatisfaction, and even fines or waste collection rejection, increasing operational pressure and costs.



System Stability & Maintenance

Overflowing System Alerts:
When the smart bin detects that it is 80% full, it immediately sends an alert to the manager's dashboard and notifies the staff through their devices. Staff can then empty the bin before it overflows.

Operational Data Support

Smart Cost Tracking System:
The system analyzes waste management costs and highlights savings from improved sorting and reduced waste disposal fees.

AI Auto-Generated Reports:
Weekly and monthly operational reports, including waste volume and sorting accuracy, are automatically generated and sent to the manager's email.

Environmental Impact Dashboard:
The smart bin collects daily sorting data and calculates the environmental impact, such as reduced landfill waste and increased recycling rates.

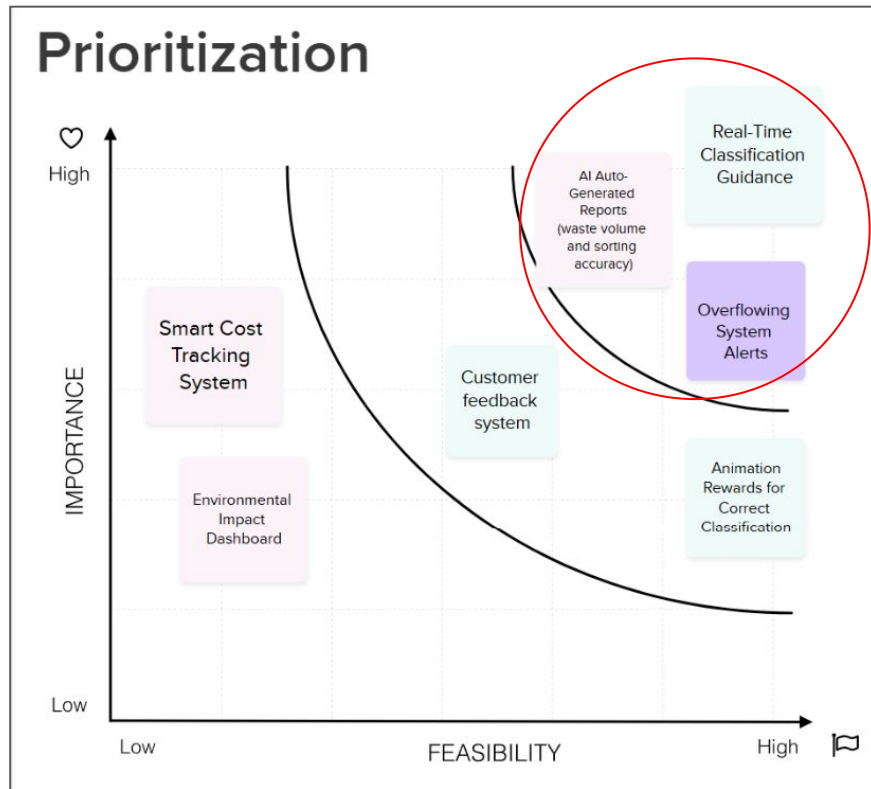
Hills

- The fast food chain manager can easily check the waste bin status from his phone and see that everything is operational and properly sorted.
- When a customer places waste on the platform, the smart bin guides them to separate leftovers and containers into the correct bins.

Smart Bin Feature Prioritization

Prioritized Feature

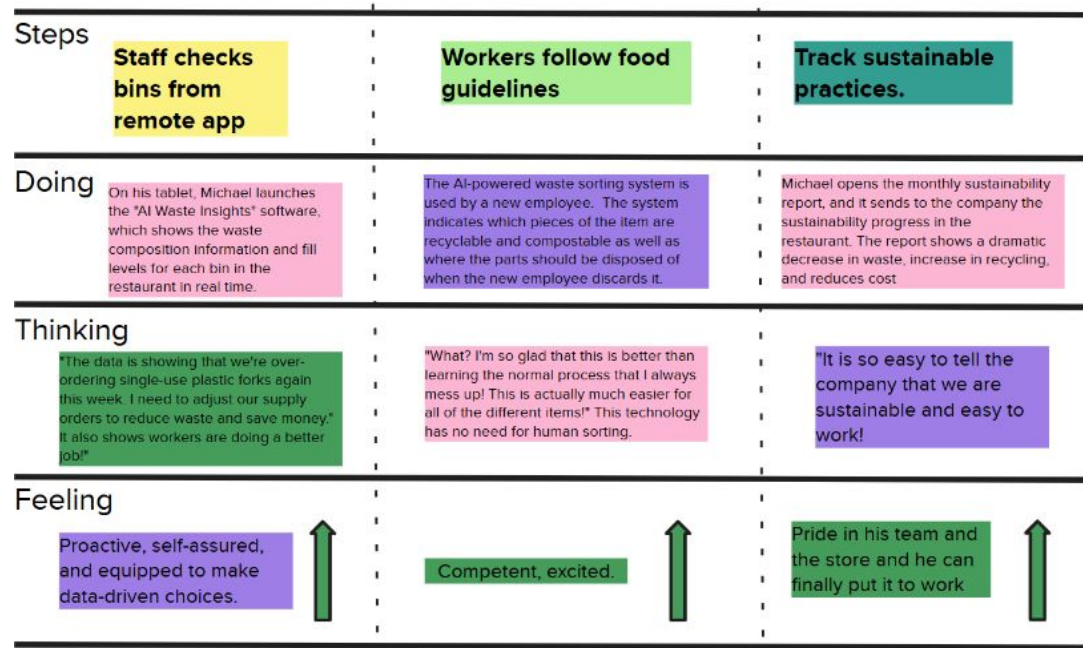
- **Real-Time Classification Guidance**
 - Uses real-time camera recognition
 - Displays on-screen instructions for correct disposal
 - Aligns with Hill: Guiding users to sort waste accurately
- **Overflowing System Alerts**
 - Sends alerts when bin is 80% full
 - Notifies staff through dashboard or mobile
 - Matches Hill: Managers can monitor bin status remotely
- **AI Auto-Generated Reports**
 - Real-time/Weekly/monthly reports on waste volume and sort accuracy
 - Sent directly to manager's email
 - Enhances monitoring and operational insights



Scenario Map - Overview

Michael and his team at a restaurant use AI-powered tools to manage waste efficiently, improve sustainability, and follow food safety guidelines all while tracking progress remotely.

TO-BE-SCENARIO MAP



Scenario Map - Key Takeaways & Benefit

Key Insights:

- Remote Monitoring: Staff can monitor waste bins from anywhere.
- AI-Driven Sorting: Reduces errors, simplifies training for new employees.
- Sustainability Reporting: Tracks progress and reduces operational costs.

Emotional Journey:

- Starts with empowerment (staff using data), moves to confidence (easy food disposal process), ends with pride (sustainability achievements shared).

Business Impact:

- Better resource management.
- Higher recycling and composting rates.
- Cost reduction through optimized supply ordering and waste handling.

Prototype Key Tasks

Customer:

1. Placing waste on the counter above the bins to determine which piece of the waste goes to which bin

Manager:

1. Monitor bin status remotely
2. Resolve a smart bin full or error notification

Usability Testing Script

Scenario Customer:

You are just finishing up your meal at the fast-food restaurant, you are about to take the waste and leftovers to the waste bins.

1. Are you going to segregate the waste?
2. Do you know how the wastes are segregated?
3. How long is it going to take just to dispose the waste?

Usability Testing Script

Scenario Manager:

You are a store manager of a fast-food chain. When it comes to cleaning out the waste bins, it is a big hassle. Customers often don't follow the rule of segregating wastes and it hard to know if the bins are full and need cleaning.

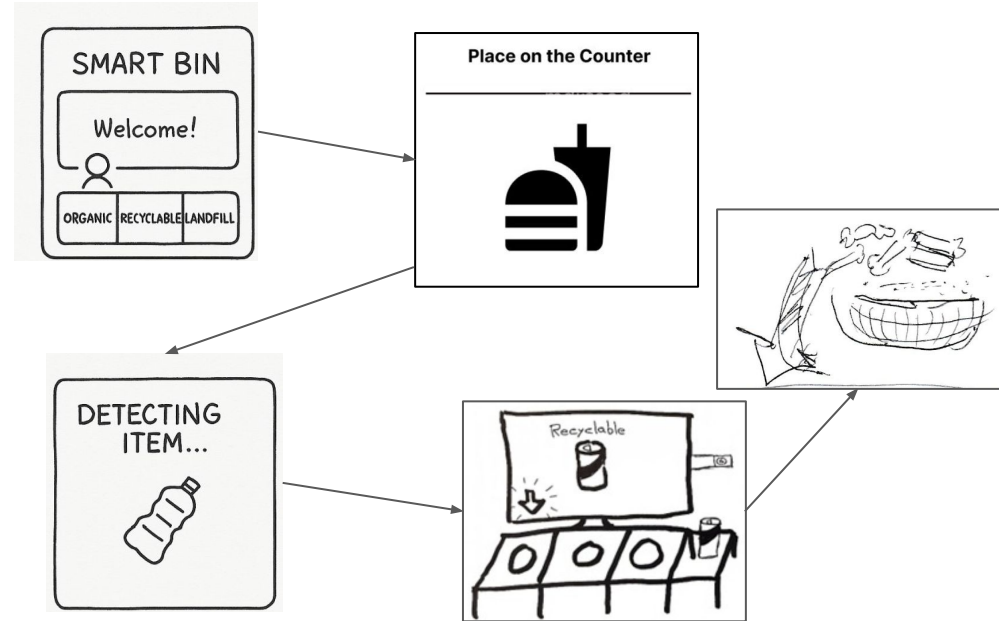
1. During the busy hours, you need to know if the waste bins are overflowing, but you don't have the time to frequently check them.
2. When cleaning out the waste, a large portion of the wastes are not sorted properly, and the need to be segregated again.

Low-Fidelity Prototype

Scenario Customer: Placing waste on the counter above the bins to determine which piece of the waste goes to which bin

Task 1:

You want to start disposing of waste into the bin. How can you use the prototype to ensure it's properly disposed of?



Place on the Counter



Cup



No, Not cup



Recycle

What is it?

Fries
Container



Plastic
Bottle



Metal
Spoon



Other



THANK YOU
Have a good day

Low-Fidelity Prototype

Scenario Manager: Monitor Bin Status & Resolve a smart bin full or error notification

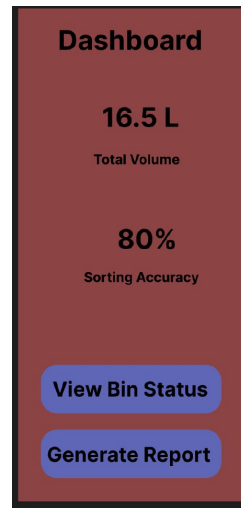
Task 2:

You want to quickly check which bins in your store are close to overflowing or have classification issues. How would you do that using the dashboard?”

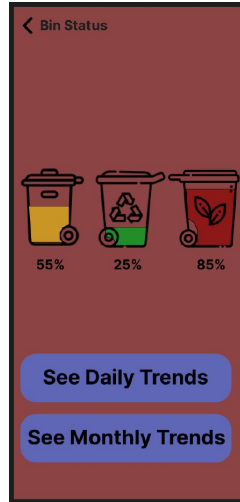
Task 3:

You want to receive a notification when one of your bins has a sorting issue, and resolve the issue promptly at the right time and location. How can you use this prototype?

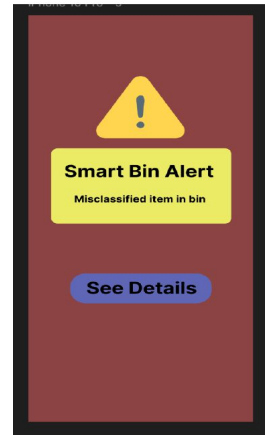
Screen 1



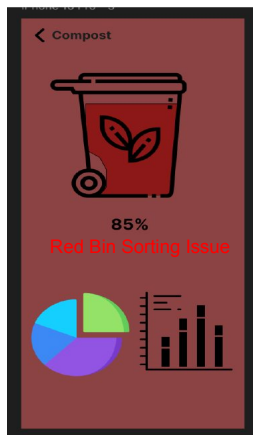
Screen 2



Screen 3



Screen 4



Dashboard

16.5 L

Total Volume

80%

Sorting Accuracy

View Bin Status

Generate Report

< Bin Status



See Daily Trends

See Monthly Trends



Smart Bin Alert

Misclassified item in bin

See Details

< Compost



85%

Red Bin Sorting Issue



Conclusion

Reimagining Waste Management with AI

- AI-powered bin simplifies waste sorting in restaurants
- Enables remote monitoring and real-time insights
- Reduces errors and supports sustainability goals
- Boosts staff confidence and encourages better habits
- Scalable, user-friendly, and impactful solution

Next Steps:

- Enhance user feedback mechanisms
- Pilot in multiple store locations
- Integrate rewards into mobile apps to further drive behavioral change